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# Moral reasoning development: norms for Defining Issue Test-2 (DIT2)

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**ABSTRACT**

This article presents normative information for the Defining Issue Test Version 2 (DIT2) schema scores and most common summary scores based on secondary data from the last 10 years DIT2 database (N = 73740, Mage = 23.11, SD = 7.87, the age range in year = 12–95) maintained by the University of Alabama's Center for the Study of Ethical Development from 2011 to 2020. More specifically, the study provides (1) norms by education; (2) norms by gender and education; and (3) norms by gender and age for the DIT2. The norms indicate a general but not consistent trend that moral reasoning scores increase by education level. Additionally, the Personal Interest and Maintain Norms scores are higher for males than females for all education levels and age groups while Postconventional, N2, and Type indicator scores are higher for females than males for all education levels and age groups.

**KEYWORDS**

Defining Issues Test 2 (DIT2); moral judgment; stage; norms

## INTRODUCTION

Moral development was first described by Piaget (1932) and then developed and studied by Kohlberg (1969). Kohlberg signifies changes in a person's form or structure of ideas rather than just an increase in understanding of cultural values that typically results in ethical relativism. Kohlberg's (1984) theory is predicated on recognition of the understanding that morality is both cognitive and developmental. Thus, this theory of moral judgment development (Kohlberg, 1958, 1969) includes that people progress through six stages of moral judgment (Table 1, Rest, 1979; van den Enden, 2019). Each stage describes a different belief regarding moral issues. Individuals are expected to move through these stages as they steadily mature from simpler to more complex moral thinking (Thoma & Dong, 2014).

Rest, Thoma, et al. (1999a) presented the Defining Issues Test (DIT) by employing a neo-Kohlbergian approach to morality. The neo-Kohlbergian view builds on Kohlberg's early work and corrects some of its weaknesses. Central to this approach is the Four Component Model (J. Rest & Narvaez, 1994; M. Bebeau, 1994; M. J. Bebeau, 2002) whereby moral functioning is understood to take place through the interaction of four processes: (1) moral sensitivity; (2) moral judgment; (3) moral motivation; and (4) moral character. Moral sensitivity involves recognizing moral situations and requirements (J. Rest, 1986). Moral motivation includes prioritizing and pursuing choices made over other options, and moral character involves effecting actions and responses deemed to be good (J. R. Rest, 1994; J. Rest, 1986).

DIT is concerned with moral judgment – the capacity to work out what to do and why in circumstances when a moral good is at stake. For neo-Kohlbergian's moral judgment operates at three levels involving codes of conduct, intermediate concepts, and bedrock schemas. Codes

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**Table 1.** Stages of moral judgment (retrieved from "van den Enden et al., 2019," p.2. and originally based on Rest, 1979).

| Stage | Description |
|---|---|
| 1 | Focus on justice in terms of power and obedience; the right thing to do is what leads to the least punishment. |
| 2 | Focus on direct or indirect satisfaction of one's own needs. Fulfilling the needs of others is based on principles of reciprocity; if I am good to someone else, this person will later also be good to me. |
| 3 | Focus on what is approved by (significant) others. Intentions become important in judging behavior and therefore role taking is necessary. |
| 4 | Focus on fulfilling one's duty in society, obeying rules and showing respect for authority; laws are justified by means of maintaining social order itself, not the underlying principles or rights they are to protect. |
| 5 | Focus on individual autonomy, rights, personal values and opinions. Laws have a social utility and can be changed by social contract. |
| 6 | Focus on abstract universal principles such as the categorical imperative. Although these can be derived from other sources, individuals actively have to choose to live upon such a principle themselves. |

of conduct are most specific and rigid, requiring limited comprehensive interpretation (Thoma, 2006), whereas intermediate concepts refer to the application of concepts, such as honesty and justice, etc., to specific contexts, and as such need to be interpreted and self-constructed (Thoma, 2006). Although DIT is associated with all three levels of moral judgment, DIT scores are understood to directly evaluate only the most general bedrock level (schemas) of functioning (J. Rest et al., 1999a). This is the most inclusive and context-free system for evaluating moral circumstances. When more automatic and context-specific interpretative systems fail or give contradictory or insufficient information, this one is utilized as a default system. DIT scores symbolize numerous sources: directly evaluating default schema and indirectly affecting the default schema on systems at the further higher levels (Thoma, 2006). Figure 1.

Employing schemas rather than the six stages of moral development, the DIT assesses three schemas, including the Personal Interest moral schema score (Stage 2/3), Maintaining Norms moral schema score (Stage 4), and Postconventional moral schema score (P Score, Rest, Thoma & Edwards, 1997). According to Rest et al. (2000), "schemas are general knowledge structures residing in long-term memory" (p. 389). DIT or DIT1 was first published in 1974 and was designed to consider seven validity and reliability criteria, including (1) deistinction of different age/education groups, (2) longitudinal gains, (3) relationship between cognitive ability measures, (4) sensitivity to moral education interventions, (5) correlations to prosocial behavior and preferred professional decision-making, (6) predicting political preference and perspective, and (7) reliability (J. R. Rest, Narvaez, et al., 1999). Later, the Defining Issues Test 2 (DIT2) was published (J. R. Rest, Narvaez, et al., 1999). DIT-2 is a theoretically produced and empirically tested measure of moral reasoning that has been effectively used to evaluate the "psychological construct that characterizes the process by which people determine that one course of action in a particular situation is morally right and another course of action is wrong" (J. Rest et al., 1997, p. 5). In comparison with the original DIT, the dilemmas and items were updated in DIT2, the test was shortened, and instructions were clearly expressed (J. R. Rest, Narvaez, et al., 1999). An index is a number that is described as a general score by which a respondent is represented (J. Rest et al., 1997). A new developmental index (N2 to replace the P index) was designed (J. R. Rest, Narvaez, et al., 1999; J. Rest et al., 1997), and a new procedure to check for participant reliability was created, such as inspecting if fake information was given (J. R. Rest, Narvaez, et al., 1999). DIT2 includes four developmental indices (i.e., Personal Interests, Maintaining Norms, Postconventional Score, and N2 score) and several developmental profile and phase indices (e.g., Type indicator) which are described in further detail below.

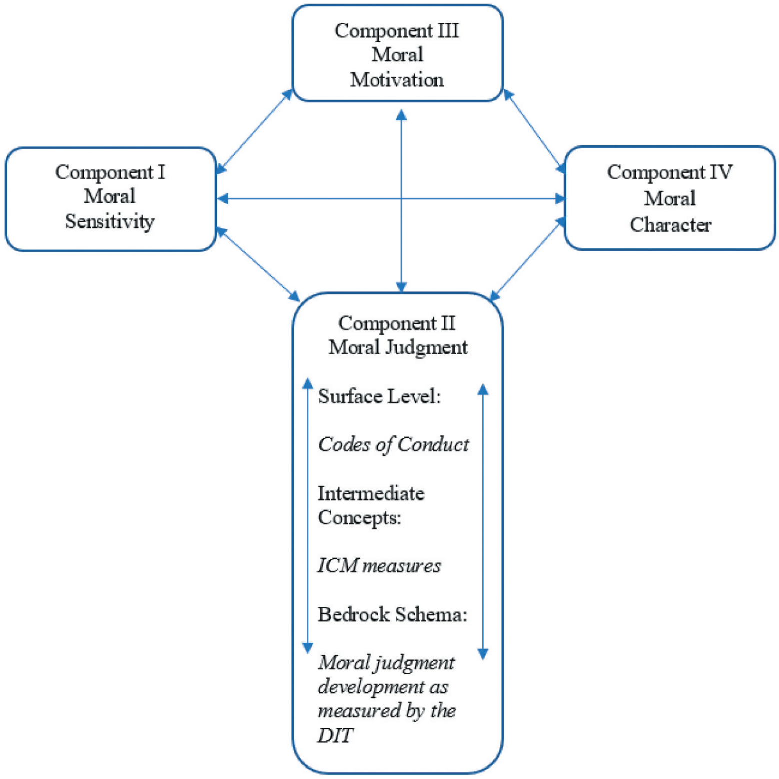


**Figure 1.** The four component model (retrieved from "Thoma, 2006," p.73 and originally based on Rest, 1983).

### *The Personal Interest moral schema score (PI)*

In the neo-Kohlbergian approach, the first schema, called the Personal Interest schema, is derived from Kohlberg's second and third stages. PI is the least developed of the three schemas, with reasoning centered on the self and those close to the self (J. Rest et al., 1999b, 2000). In other words, when examining moral dilemmas, those who employ the PI schema in moral decisions focus on how a course of action will affect the individuals involved rather than how it would affect society as a whole. In this way, a PI schema emphasizes a view that considers the rewards and failures each person may directly feel in response to a moral quandary (J. Rest et al., 1999a). As Rest et al. (2000) stated, this schema "justifies a decision as morally right by appealing to the personal stake the actor has in the consequences of an action" (p. 387). PI schema is available when respondents can reliably complete the DIT (e.g., 9th-grade reading level) and achieve PI scores ranging from 0 to 100; a higher score represents a higher level of PI reasoning (see DIT guide).

### *The Maintaining Norms moral schema score (MN)*

The Maintaining Norms schema is derived from Kohlberg's stage 4. It adopts a society-wide perspective and is more developed than PI; reasoning at this level is concerned with authority, rules, roles, and the generally accepted way of life for a community in which the individual is a member (J. Rest et al., 1999b, 2000). MN's scores range from 0 to 98; a higher score represents a higher level of MN reasoning (see DIT guide). In the MN schema, individuals place value on upholding social order, which establishes morality. According to this schema, the law serves as the cornerstone of order without which there would be anarchy, which is something responsible people would naturally wish to avoid.

### *The postconventional moral schema score (P Score)*

The Postconventional Moral Schema Score (P score) is derived from Kohlberg's Stages 5 and 6. The P score represents the relative importance each subject place on Kohlberg's postconventional level of moral development. This means that it is the most developed, with reasoning at this level focusing on shared principles that are subject to examination and debate rather than just depending on established ways of doing things (J. Rest et al., 1999b, 2000). The P score is calculated on the basis of allocating points to ranking data: four points to the stage of the highest ranked item to one point to the stage of the fourth ranked item. The P score is calculated by dividing the total number of points across the six dilemmas for postconventional items (hence the "P") by the total number of points. The P score can range from 0% to 95% (not 100% because not every dilemma has four possible postconventional items). Different from the MN schema, where the endeavor to reach moral consensus is based on appealing to regular practice and existing authority, the attempt to reach moral consensus in this schema is accomplished through appealing to ideals and logical coherence (Auger & Gee, 2016).

### *The N2 index*

A new index (N2) for moral reasoning has been introduced in the DIT2, replacing the prior index, P. The N2 index is based on (a) the extent to which the subject ranks postconventional items above PI and MN and (b) the difference in ratings of PI from the P score. These parts of N2 are integrated into one score for each participant by adding the postconventional score to the rating data weighted by three. The advantage of the N2 index is – as a single index – it can account for those who heavily emphasize the importance of Postconventional items and deemphasize the importance of PI and MN items. Thus, those with high N2 scores would therefore have high P scores but low MN and PI scores. Because N2 scores are modified to have the same mean and standard deviation as the P score, comparisons between P score and N2 can also be conducted. Different from other DIT indices that use only ranking data, the N2 index is calculated based on both ratings and rankings to measure moral judgment without asking for additional work from respondents. N2 also has more rigid rules for managing missing data than the P score (J. Rest et al., 1997). Thus, more protocols are invalidated for missing data in the N2 index than for the postconventional score. Similar to the P score, N2 ranges from 0 to 95; a higher score represents a higher level of N2 reasoning. It is a percentage score but does not quite reach 100 because there are not the same number of item types for each of the dilemmas (see DIT guide).

### *Type indicator*

Type Indicator is a developmental profile and phase indices that provide more fine-grained methods of investigating the influences of development and/or the impact of educational interventions. (Thoma & Rest, 1999). Thoma and Rest (1999) realized that there was no clear difference between two or more schema-typed items for some participants. It was accepted as an indicator of developmental disequilibrium. Thus, criteria were determined to classify profiles as either consolidated students or transitional. So, it is possible to anticipate seven characteristics of students in different type levels; "Type I"-predominant in PI schema and consolidated; "Type 2"-predominant in PI schema, but transitional; "Type 3"-predominant in MN schema, but transitional; PI secondary schema; "Type 4"-predominant in MN schema and consolidated; "Type 5"-predominant in MN schema and transitional; postconventional secondary schema; "Type 6"-predominant in postconventional schema, but transitional, and "Type 7"-predominant in postconventional schema and consolidated (see Figure 2). It can be anticipated a person moves from consolidated profiles to transitional profiles as their development advances over the course of their lifetime, with accompanying changes in schema preference.

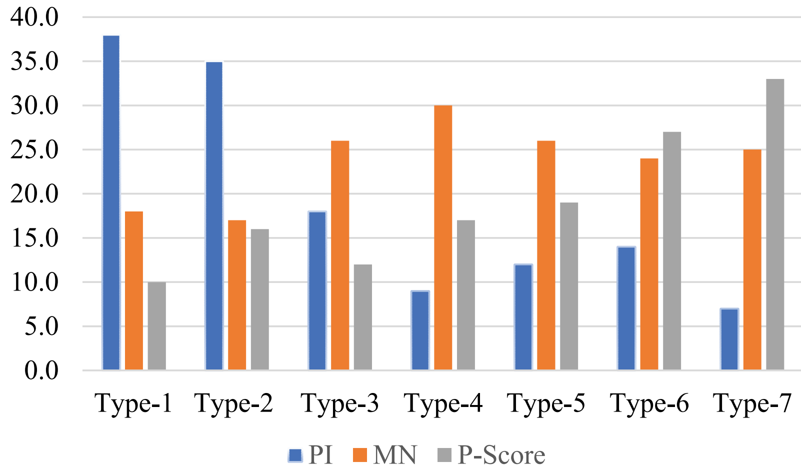


**Figure 2.** Mean DIT item ratings as a function of moral type (retrieved from "DIT2 Guide," p.21).

### *DIT2 norms*

Bebeau, Maeda, and Tichy-Reese (2003) first attempted to generate normative information for DIT2 schema scores from data sets ($N = 13386$) scored by the Center for the Study of Ethical Development since 1998. They reported a general difference between males and females by educational level as women scored higher than their male counterparts when the education level increased. The second attempt to generate normative information for DIT2 schema scores included data from 2005 to 2009 ($N = 53261$) and was done by the Center for the Study of Ethical Development in 2010. Similarly, a significant gender difference favoring females in N2 scores and P scores by level of education was reported.

Since 2018, online DIT2 has been available and became more popular and preferred over the paper version. This major change in test taking conditions (i.e., paper vs. online) supports a need for updating DIT2 norms. Like previous norms, age, gender, and education on reporting summary and schema scores, are the current focus. However, unlike previous norms, we cover a 10-year period, and offer DIT2 summary and schema scores based on age period. The current norms include a larger sample than for previous norms and are different, too, for covering a period of time associated with reduced educational gaps between men and women (Le & Nguyen, 2020) and reduced gender role differences in society (Ayisi & Krisztina, 2022)). A related expectation is that response differences for DIT2 based on gender and education will be less than in previous years.

## MATERIALS AND METHODS

### *Procedure*

In order to update normative information for DIT2 schema scores (i.e., PI, MN, and P score) and most common summary scores (i.e., N2 score and Type indicator) we used secondary data from the DIT2 database maintained by the University of Alabama's Center for the Study of Ethical Development from 2011 to 2020 (10 years). (For brief characterizations of each index, see Guide for DIT-2, p. 18–21). DIT2 has been available to researchers exclusively through the Center for the Study of Ethical Development since 1998. The database in the Center ensures consistency in data scoring and reporting and includes a satisfactory number of participants' responses from various ages, gender, and educational levels.

## Participants

### Inclusion and exclusion criteria

Data sets were selected if respondents passed the reliability check and completed 5 stories in DIT2 ($N$ = 153127). More specifically, the purged sample includes those cases that did not pass DIT2 validity checks (J. Rest, 1986), which are intended to recognize participants who did not complete the questionnaire seriously or understand its instructions properly. In this dataset, participants who had missing information, who did not provide demographic information (e.g., age, gender, education, political liberalism, citizenship, and language), and who provided mistaken information (e.g., typing 1553 for age) were eliminated from the analyses ($N$ = 90601). Only those who indicated that they were citizens of the U. S. A. and indicated that English was their primary language were included in the analyses. Finally, $N$ = 73740 participants in DIT2 data sets met these criteria and were considered as valid data sets to generate the norms. They represent diversity in age ($M_{age}$ = 23.11, $SD$ = 7.87, age range in year = 12–95), gender, education, and political liberalism. Regarding the DIT2 type, 53.5% of the participants completed the paper-based version of the test (see Table 2).

Level of Education was completed as the highest level of formal education attained, if participants are currently working at that level (e.g., freshman in college) or if they have completed that level (e.g., if you finished your freshman year but have gone on no further). There were 12 categories from Grade 1 to Doctoral Degree. The level of political views was completed with a 5-point Likert-type scale from very liberal to very conservative.

### Data collection instrument

***THE DEFINING ISSUES TEST 2 (DIT2;* Rest, Narvaez, Thoma, & Bebeau, 1999a.** The DIT2 is a quantitative instrument designed to measure an individual's level of moral reasoning. The instrument most often used to operationalize moral development and place individuals within a level of moral reasoning is Rest's Defining Issues Test (DIT). DIT2 assesses recognition, comprehension, and preference. Participants complete the DIT2 by reading the story and then choosing on a 3-point scale what the protagonist should do (e.g., on the Heinz dilemma the choices are as follows: (1) steal the drug, (2) not steal, or (3) can't decide). Each story is followed by 12 statements (items) concerning the specific moral issue. It is a multiple-choice instrument that yields a continuous number. For each

**Table 2.** Demographic descriptive statistics of participants.

| Variables<br>Age (year) | Participants' profile<br>Adolescence and Adulthood | Mean±SD or % (n)<br>23.111 ± 7.87 |
|---|---|---|
| Gender | Male | 49.7 (36678) |
| | Female | 50.3 (37062) |
| Education | Grade 1–6 | 1.1 (781) |
| | Grade 7–9 | 2.2 (1614) |
| | Grade 10–12 | 7.2 (5317) |
| | Voc/Tech | 1.2 (903) |
| | Jr. College | 3.8 (2776) |
| | Freshman | 25.7 (18985) |
| | Sophomore | 9.7 (7116) |
| | Junior | 11.8 (8751) |
| | Senior | 20.2 (14894) |
| | Prof. degree | 8.8 (6490) |
| | MS degree | 3.3 (2441) |
| | Doctoral degree | 5.0 (3672) |
| Political Liberalism | Very liberal | 7.4 (5423) |
| | Somewhat liberal | 23.9 (17646) |
| | Neither liberal nor conservative | 27.8 (20518) |
| | Somewhat conservative | 30.8 (22748) |
| | Very conservative | 10.1 (7405) |
| DIT-2 Type | Online | 46.5 (34285) |
| | Paper | 53.5 (39455) |

$N$ = 73740, Voc/Tech: Vocational/Technical, Jr: Junior, Prof: Professional, MS: Masters.

story, participants first rated all 12 items on a 5-point scale ranging from 1 (*no importance*) to 5 (*great importance*) and subsequently ranked which four items they found most important. This procedure is repeated for the other stories (Thoma, 2006). DIT2 has improved validity, enhanced input reliability checks, and yields better trends than DIT1 (J. R. Rest, Narvaez, et al., 1999, p. 8). The DIT-2 consists of five short moral stories that aim to understand how respondents define the issues in a social problem:

(1) Famine: What should Mustaq Singh do? Do you favor the action of taking food?
(2) Reporter: Do you favor the action of reporting the story?
(3) School Board: Do you favor calling off the next Open Meeting?
(4) Cancer: Do you favor the action of giving more medicine?
(5) Demonstration: Do you favor the action of demonstrating in this way?

The DIT2 protects against participants' inconsistency by having meaningless items within the 12 statements for each scenario. If a participant rates or ranks these items with importance, this reflects that the participant might not be able to differentiate this item from the relevant statements that represent the larger moral argument and that they might be trying to fake a high score (O'Connor, 2018).

## DATA ANALYSIS

The data were analyzed in the SPSS (26.0) program. Means, standard deviations, percentiles (i.e., 25 and 75%iles), and median were reported for each schema and summary score. The study provides (1) norms by education, (2) norms by gender and education, and (3) norms by gender and age for the DIT2. Though these data represent some diversity in age, gender, education levels, political liberalism, citizenship, and language, researchers should use caution in comparing their data sets with the normative information – the provided information acts as a general guide for comparisons of their data sets with the larger sample.

## RESULTS

DIT2 scores for each of the moral schemas by level of education are shown in Table 3. This includes DIT2 mean scores, standard deviations, percentiles, and median for each of the moral schemas (i.e., PI, MN, and P score) by level of education, from 2011 to 2020.

**Table 3.** DIT2 means, standard deviation, and percentiles for schema scores by educational level for respondents.

| | | Schema Scores | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Personal Interest (PI) | | | | | Maintain Norms (MN) | | | | | Postconventional (P score) | | | | |
| Edu. Level | N | M | SD | 25% | Med | 75% | M | SD | 25% | Med | 75% | M | SD | 25% | Med | 75% |
| Grade 1–6 | 781 | 24.6 | 11.8 | 16 | 24 | 32 | 34.5 | 13.9 | 24 | 34 | 44 | 35.6 | 15.7 | 24 | 36 | 46 |
| Grade 7–9 | 1614 | 27.3 | 13.4 | 18 | 26 | 36 | 33.3 | 14.4 | 22 | 32 | 44 | 32.5 | 16.7 | 20 | 32 | 44 |
| Grade 10–12 | 5317 | 27.5 | 13.0 | 18 | 26 | 36 | 33.5 | 13.9 | 24 | 34 | 44 | 32.8 | 16.4 | 20 | 32 | 44 |
| Voc/Tech | 903 | 27.1 | 13.0 | 18 | 26 | 36 | 36.8 | 13.8 | 28 | 36 | 46 | 29.2 | 14.8 | 18 | 28 | 40 |
| Jr. College | 2776 | 28.5 | 12.7 | 20 | 28 | 38 | 34.6 | 13.6 | 24 | 34 | 44 | 30.4 | 15.5 | 18 | 28 | 42 |
| Freshman | 18985 | 28.1 | 12.7 | 18 | 28 | 36 | 34.8 | 13.5 | 24 | 34 | 44 | 31.1 | 15.3 | 20 | 30 | 42 |
| Sophomore | 7116 | 26.1 | 12.8 | 18 | 26 | 36 | 33.7 | 13.9 | 24 | 34 | 44 | 34.4 | 15.7 | 22 | 34 | 46 |
| Junior | 8751 | 25.8 | 12.8 | 16 | 24 | 34 | 33.9 | 14.0 | 24 | 34 | 44 | 34.6 | 15.8 | 22 | 34 | 46 |
| Senior | 14894 | 24.1 | 12.5 | 14 | 22 | 32 | 33.3 | 14.2 | 22 | 32 | 44 | 37.2 | 16.0 | 26 | 36 | 48 |
| Prof. degree | 6490 | 21.2 | 11.6 | 12 | 20 | 28 | 31.9 | 14.6 | 20 | 32 | 42 | 42.3 | 16.0 | 30 | 42 | 54 |
| MS degree | 2441 | 22.7 | 12.3 | 14 | 22 | 30 | 33.8 | 14.9 | 22 | 32 | 44 | 38.5 | 15.6 | 28 | 38 | 50 |
| Doc. degree | 3672 | 23.4 | 11.9 | 14 | 22 | 32 | 32.3 | 14.2 | 22 | 32 | 42 | 38.9 | 16.0 | 28 | 38 | 50 |

*N* = 73740, M=Mean, *SD*=Standard Deviation, Med: Median Edu: Education, Voc/Tech: Vocational/Technical, Jr: Junior, Prof: Professional, MS: Masters, Doc: Doctoral.

**Table 4.** DIT2 means, standard deviation, and percentiles for summary scores by educational level for respondents.

| | Summary Scores | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | N2 Score | | | | | | Type Indicator | | | | | |
| Edu. Level | N | M | SD | 25% | Med | 75% | N | M | SD | 25% | Med | 75% |
| Grade 1–6 | 781 | 35.45 | 14.74 | 25.3 | 36.1 | 45.6 | 780 | * | * | 4 | 5 | 6 |
| Grade 7–9 | 1613 | 31.50 | 16.64 | 18.8 | 31.2 | 44.1 | 1611 | * | * | 3 | 4 | 6 |
| Grade 10–12 | 5314 | 31.82 | 16.27 | 19.1 | 31.5 | 44.1 | 5312 | * | * | 3 | 4 | 6 |
| Voc/Tech/Jr. | 903 | 28.05 | 14.90 | 16.6 | 26.2 | 37.8 | 899 | * | * | 3 | 4 | 6 |
| Jr. College | 2774 | 29.08 | 15.49 | 16.8 | 27.6 | 40.6 | 2775 | * | * | 2 | 4 | 6 |
| Freshman | 18976 | 30.26 | 15.19 | 18.5 | 29.8 | 41.3 | 18966 | * | * | 3 | 4 | 6 |
| Sophomore | 7111 | 33.67 | 15.40 | 22.2 | 33.5 | 44.9 | 7110 | * | * | 3 | 5 | 6 |
| Junior | 8750 | 34.03 | 15.57 | 22.2 | 34.0 | 45.7 | 8742 | * | * | 3 | 4 | 6 |
| Senior | 14887 | 36.96 | 15.68 | 25.6 | 37.5 | 48.5 | 14880 | * | * | 3 | 5 | 7 |
| Prof. degree | 6486 | 42.41 | 14.82 | 32.3 | 43.7 | 53.4 | 6430 | * | * | 4 | 6 | 7 |
| MS degree | 2440 | 38.38 | 15.17 | 28.0 | 39.1 | 49.6 | 2440 | * | * | 4 | 5 | 7 |
| Doc degree | 3672 | 38.47 | 15.10 | 27.9 | 39.0 | 49.2 | 3636 | * | * | 4 | 6 | 7 |

*N* = 73740, M=Mean, *SD*=Standard Deviation, Med: Median, Edu: Education, Voc/Tech: Vocational/Technical, Jr: Junior, Prof: Professional, MS: Masters.
*No M and SD were reported since Type Indicator has ordinal responses.

Summary scores by the level of education for DIT2 are also shown in Table 4. This includes DIT2 mean scores, standard deviations, percentiles, and median for N2 summary score while percentiles and median for Type indicator score by level of education, from 2011 to 2020.

### College class year

Because a significant portion of the current DIT-2 data was from college students, we investigated whether there are significant college class year differences in schema scores (i.e., PI, MN, and P score) and N2 summary score. A series of one-way analysis of variance (ANOVA) was performed to assess whether schema and summary scores could be differentiated by college class year (i.e., Freshman, Sophomore, Junior, and Senior). The results showed that there was a statistically significant difference between groups as determined by one-way ANOVA for PI ($F(3, 49740) = 288.686$, $p < .001$, $\eta2 = .017$), MN ($F(3, 49740) = 35.728$, $p < .001$, $\eta2 = .002$), P score ($F(3, 49740) = 35.728$, $p < .001$, $\eta2 = .025$), ad N2 score ($F(3, 49740) = 35.728$, $p < .001$, $\eta2 = .031$). The Tukey post hoc tests revealed that seniors received significantly lower PI and MN and significantly higher P scores and N2 than freshmen ($p < .001$) (Tables 3, 4).

### Gender differences by education levels

We investigated whether there are gender differences in schema scores (i.e., PI, MN, and P score) and N2 summary score that are qualified by the education level. To be able to report more concise results, we combined the educational level into four subcategories (i.e., (1) Grade 1–12, (2) Voc/Tech and/or Jr, (3) Undergraduate Degree, and (4) Graduate or Prof Degree). A series of independent $2 \times 4$ full factorial ANOVA was performed to assess whether schema and summary scores could be predicted by gender (i.e., female and male), education level (i.e., Grade 1–12, Vocational/Technical or Junior (Voc/Tech or Jr), Undergraduate, and Graduate or Professional Degree), and the interaction of gender and education level.

The first ANOVA results showed significant main effects for education level, ($F(3, 73728) = 418.641$, $p < .001$; $\eta2 = .017$); gender, ($F(1, 73728) = 243.273$, $p < .001$; $\eta2 = .013$), and a significant interaction, ($F(3, 73728) = 5.027$, $p < .001$; $\eta2 = .001$) for PI score. The significant interaction showed that the relationship between PI score and education level depends on gender. Simple main effects test indicated that if there was a significant difference between PI scores of males and females at each education level $p < .001$, males, especially those from lower education levels, received higher PI scores than females. Post-hoc analysis indicated significantly different mean PI scores between all education levels (Tukey HSD, $p = .001$ for Grade 1–12 and Voc/Tech or Jr and $p < .001$ for the rest of the groups).

The second ANOVA results showed significant main effects for education level, ($F(3, 73728) = 56.987$, $p < .001$; $\eta 2 = .002$); gender ($F(1, 73728) = 110.268$, $p < .001$; $\eta 2 = .001$), and a significant interaction, ($F(3, 73728) = 6.722$, $p < .001$; $\eta 2 = .001$) for MN score. The significant interaction showed that the relationship between MN score and education level depends on gender. Simple main effects test indicated that there was a significant difference between MN scores of males and females at each education level $p < .001$; males, especially those from lower education levels, received higher MN scores than females. Post-hoc analysis indicated significantly different mean MN scores between all education levels (Tukey HSD, $p = .014$ for Grade 1–12 and Undergraduates and $p < .001$ for the rest of the groups).

The third ANOVA results showed significant main effects for education level, ($F(3, 73728) = 692.413$, $p < .001$; $\eta 2 = .027$); gender ($F(1, 73728) = 557.459$, $p < .001$; $\eta 2 = .008$), and a significant interaction, ($F(3, 73728) = 9.182$, $p < .001$; $\eta 2 = .001$) for the P score. The significant interaction showed that the relationship between the P score and education level depends on gender. Simple main effects test indicated that there was a significant difference between the P scores of males and females at each education level $p < .001$; females, especially those from higher education levels, received higher P scores than males. Post-hoc analysis indicated significantly different mean P scores between all education levels (Tukey HSD, $p < .001$ for all groups).

The fourth ANOVA results showed significant main effects for education level, ($F(3, 73699) = 868.870$, $p < .001$; $\eta 2 = .034$); gender ($F(1, 73699) = 490.250$, $p < .001$; $\eta 2 = .007$), and a significant interaction, ($F(3, 73699) = 12.787$, $p < .001$; $\eta 2 = .001$) for N2 score. The significant interaction showed that the relationship between N2 score and education level depends on gender. Simple main effects test indicated that there was a significant difference between N2 scores of males and females at each education level $p < .001$; females, especially those from higher education levels, received higher P scores than males. Post-hoc analysis indicated significantly different mean N2 scores between all education levels (Tukey HSD, $p < .001$ for all groups) (Table 5).

### *Gender differences by age period*

We also investigated whether there are gender differences in schema scores (i.e., PI, MN, and P score) and N2 summary score that are qualified by age. To report more concise results, we split participants into three age periods (i.e., (1) adolescence (12–18-year-old), (2) young adulthood (19–30-year-old), and (3) middle-late adulthood (31–95-year-old)). A series of independent $2 \times 3$ full factorial ANOVA was performed to assess whether schema and summary scores could be predicted by gender (i.e., female and male), age period (i.e., adolescence, young adulthood, and middle-late adulthood), and the interaction of gender and age period. For ordinal Type Indicator score, we performed Kendall's tau-b ($\tau$b) correlation.

The first ANOVA results showed significant main effects for age period, ($F(2, 73730) = 383612$, $p < .001$; $\eta 2 = .010$); gender ($F(1, 73730) = 367.415$, $p < .001$; $\eta 2 = .005$), and a significant interaction, ($F(2, 73730) = 12.837$, $p < .001$; $\eta 2 = .001$) for PI score. The significant interaction showed that the relationship between PI score and age groups depends on gender. Simple main effects test indicated that there was a significant difference between PI scores of males and females at each age period $p < .001$; males, especially those from younger age groups, received higher PI score than females. Post-hoc analysis indicated significantly different mean PI scores between all age periods (Tukey HSD, $p < .001$).

The second ANOVA results showed significant main effects for age period, ($F(2, 73730) = 188.723$, $p < .001$; $\eta 2 = .005$); gender ($F(1, 73730) = 90.598$, $p < .001$; $\eta 2 = .001$), and a significant interaction, ($F(2, 73730) = 15.974$, $p < .001$; $\eta 2 = .001$) for MN score. The significant interaction showed that the relationship between MN score and age groups depends on gender. Simple main effects test indicated that there was a significant difference between MN scores of males and females at young adulthood

**Table 5.** Means, standard deviation, and percentiles for schema and summary scores by educational level and gender for respondents.

| | Male | | | | | | Female | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Edu. Level | N | M | SD | 25% | Med | 75% | N | M | SD | 25% | Med | 75% |
| **Personal Interest (PI)** | | | | | | | | | | | | |
| Grade 1–12 | 4325 | 28.5 | 12.9 | 18 | 28 | 38 | 3385 | 25.6 | 13.0 | 16 | 24 | 34 |
| Voc/Tech or Jr | 1866 | 28.8 | 13.0 | 20 | 28 | 38 | 1813 | 27.5 | 12.5 | 18 | 28 | 36 |
| Undergraduate | 24766 | 27.5 | 12.8 | 18 | 26 | 36 | 24978 | 24.9 | 12.6 | 16 | 24 | 34 |
| Graduate or Prof. | 5923 | 23.3 | 12.1 | 14 | 22 | 32 | 7162 | 21.3 | 11.6 | 12 | 20 | 28 |
| **Maintain Norms (MN)** | | | | | | | | | | | | |
| Grade 1–12 | 4325 | 34.6 | 13.8 | 24 | 34 | 44 | 3385 | 32.2 | 14.2 | 22 | 32 | 42 |
| Voc/Tech or Jr | 1866 | 36.0 | 14.0 | 26 | 36 | 46 | 1813 | 34.3 | 13.3 | 24 | 34 | 44 |
| Undergraduate | 24766 | 34.5 | 13.9 | 24 | 34 | 44 | 24978 | 33.5 | 13.9 | 24 | 34 | 44 |
| Graduate or Prof. | 5923 | 33.2 | 15.0 | 22 | 32 | 44 | 7162 | 31.7 | 14.1 | 20.8 | 30.4 | 42 |
| **Postconventional (P score)** | | | | | | | | | | | | |
| Grade 1–12 | 4325 | 30.4 | 15.6 | 18 | 30 | 42 | 3385 | 36.2 | 16.7 | 24 | 36 | 48 |
| Voc/Tech or Jr | 1866 | 28.6 | 14.8 | 18 | 26 | 38 | 1813 | 31.7 | 15.6 | 20 | 30 | 42.2 |
| Undergraduate | 24766 | 32.0 | 15.5 | 20 | 32 | 42 | 24978 | 35.9 | 16.0 | 24 | 36 | 48 |
| Graduate or Prof. | 5923 | 38.3 | 16.1 | 26 | 38 | 50 | 7162 | 42.2 | 15.8 | 32 | 42 | 54 |
| **N2 Score** | | | | | | | | | | | | |
| Grade 1–12 | 4324 | 29.6 | 15.5 | 17.8 | 29.2 | 40.8 | 3384 | 35.3 | 16.5 | 22.3 | 36.1 | 48 |
| Voc/Tech or Jr | 1864 | 27.5 | 14.9 | 16.1 | 26 | 38.1 | 1813 | 30.2 | 15.6 | 17.8 | 29.2 | 42.1 |
| Undergraduate | 24759 | 31.6 | 15.3 | 19.8 | 31.2 | 42.7 | 24965 | 35.2 | 15.7 | 23.2 | 35.6 | 47.1 |
| Graduate or Prof. | 5921 | 38.4 | 15.2 | 27.8 | 39.4 | 50 | 7159 | 41.8 | 14.9 | 32.2 | 43 | 53.1 |
| **Type Indicator** | | | | | | | | | | | | |
| Grade 1–12 | 4319 | * | * | 3 | 4 | 6 | 3384 | * | * | 3 | 5 | 7 |
| Voc/Tech or Jr | 1864 | * | * | 2 | 4 | 6 | 1810 | * | * | 3 | 4 | 6 |
| Undergraduate | 24743 | * | * | 3 | 4 | 6 | 24955 | * | * | 3 | 5 | 7 |
| Graduate or Prof. | 5892 | * | * | 4 | 6 | 7 | 7095 | * | * | 4 | 6 | 7 |

*N* = 73740, M=Mean, *SD*=Standard Deviation, Med: Median, Edu: Education, Voc/Tech: Vocational/Technical, Jr: Junior, Prof: Professional.

*No M and SD were reported since Type Indicator has ordinal responses.

and middle-late adulthood levels $p < .001$, while non-significant difference at adolescence level $p = .090$. Males received higher MN score than females at each age period, but this difference was non-significant in adolescence. Post-hoc analysis indicated significantly different mean MN scores between all age periods (Tukey HSD, $p < .001$). MN score was highest in middle and late adulthood; however, it was higher in adolescence than in young adulthood.

The third ANOVA results showed significant main effects for age period, ($F(2, 73730) = 450.146$, $p < .001$; $\eta2 = .012$); gender ($F(1, 73730) = 696.095$, $p < .001$; $\eta2 = .009$), and a significant interaction, ($F(2, 73730) = 7.584$, $p = .001$; $\eta2 = .001$) for the P score. The significant interaction showed that the relationship between the P score and age groups depends on gender. Simple main effects test indicated that there was a significant difference between the P scores of males and females at each age period $p < .001$; females, especially those from older age groups, received higher P scores than males. Post-hoc analysis indicated significantly different mean P scores between all age periods (Tukey HSD, $p < .001$). The P score was highest in young adulthood; however, it was higher in middle and late adulthood than in adolescence.

The fourth ANOVA results showed significant main effects for age period, ($F(2, 73701) = 588.228$, $p < .001$; $\eta2 = .016$); gender ($F(1, 73701) = 604.434$, $p < .001$; $\eta2 = .008$), and a significant interaction, ($F(2, 73701) = 7.260$, $p = .001$; $\eta2 = .001$) for N2 score. The significant interaction showed that the relationship between N2 score and age groups depends on gender. Simple main effects test indicated that there was a significant difference between N2 scores of males and females at each age period $p < .001$; females, especially those from older age groups, received higher N2 scores than males. Post-hoc analysis indicated significantly different mean N2 scores between all age periods (Tukey HSD, $p < .001$). N2 score was highest in young adulthood; however, it was higher in middle and late adulthood than in adolescence (Table 6).

**Table 6.** Means, standard deviation, and percentiles for schema and summary scores by age period and gender for respondents.

| | Male | | | | | | Female | | | | | | Total | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Age Period | N | M | SD | 25% | Med | 75% | N | M | SD | 25% | Med | 75% | N | M | SD | 25% | Med | 75% |
| **Personal Interest (PI)** | | | | | | | | | | | | | | | | | | |
| Adolescence | 9095 | 29.4 | 12.7 | 20 | 29.7 | 38 | 9562 | 26.4 | 12.6 | 18 | 26 | 34 | 18657 | 27.8 | 12.8 | 18 | 28 | 36 |
| Young Adult. | 23843 | 26.5 | 12.8 | 18 | 26 | 36 | 23591 | 23.8 | 12.5 | 14 | 22 | 34 | 47434 | 25.2 | 12.7 | 16 | 24 | 34 |
| Middle-Late Adult. | 3738 | 24.6 | 12.4 | 16 | 24 | 32 | 3907 | 23.3 | 12.4 | 14 | 22 | 32 | 7645 | 24 | 12.5 | 14 | 22 | 32 |
| **Maintain Norms (MN)** | | | | | | | | | | | | | | | | | | |
| Adolescence | 9095 | 34.3 | 13.6 | 24 | 34 | 44 | 9562 | 33.9 | 13.6 | 24 | 34 | 44 | 18657 | 34.1 | 13.6 | 24 | 34 | 44 |
| Young Adult. | 23843 | 34.0 | 14.0 | 24 | 34 | 44 | 23591 | 32.4 | 13.8 | 22 | 32 | 42 | 47434 | 33.2 | 13.9 | 22.2 | 32 | 42 |
| Middle-Late Adult. | 3738 | 37.4 | 15.2 | 26 | 38 | 48 | 3907 | 35.5 | 14.9 | 24 | 26 | 46 | 7645 | 36.5 | 15.1 | 26 | 36 | 48 |
| **Postconventional (P score)** | | | | | | | | | | | | | | | | | | |
| Adolescence | 9095 | 30.2 | 15.2 | 18 | 30 | 40 | 9562 | 34.0 | 15.8 | 22 | 34 | 46 | 18657 | 32.2 | 15.6 | 20 | 32 | 42 |
| Young Adult. | 23843 | 33.7 | 15.8 | 22 | 32 | 44 | 23591 | 38.4 | 16.1 | 26 | 38 | 50 | 47434 | 36.1 | 16.2 | 24 | 36 | 48 |
| Middle-Late Adult. | 3738 | 31.6 | 16.1 | 20 | 30 | 42 | 3907 | 35.2 | 16.5 | 23.9 | 34 | 47.8 | 7645 | 33.4 | 16.4 | 22 | 32 | 44 |
| **N2 Score** | | | | | | | | | | | | | | | | | | |
| Adolescence | 9094 | 29.4 | 15.1 | 17.9 | 29.1 | 40.2 | 9556 | 33.1 | 15.7 | 20.8 | 33.4 | 45.1 | 18650 | 31.3 | 15.5 | 19.3 | 31.1 | 42.8 |
| Young Adult. | 23833 | 33.5 | 15.6 | 21.8 | 33.5 | 45.1 | 23581 | 37.8 | 15.7 | 26.3 | 38.6 | 49.5 | 47414 | 35.7 | 15.8 | 23.8 | 36.1 | 47.5 |
| Middle-Late Adult. | 3737 | 30.9 | 16.1 | 18.4 | 30.3 | 42.4 | 3906 | 33.8 | 16.3 | 21.6 | 33.2 | 45.8 | 7643 | 32.4 | 16.3 | 20 | 31.9 | 44.4 |
| **Type indicator** | | | | | | | | | | | | | | | | | | |
| Adolescence | 9086 | * | * | 2 | 4 | 6 | 9557 | * | * | 3 | 5 | 6 | 18643 | * | * | 3 | 4 | 6 |
| Young Adult. | 23799 | * | * | 3 | 4 | 6 | 23507 | * | * | 3 | 6 | 7 | 47306 | * | * | 3 | 5 | 7 |
| Middle-Late Adult. | 3730 | * | * | 3 | 4 | 6 | 3902 | * | * | 3 | 5 | 6 | 7632 | * | * | 3 | 4 | 6 |

*N* = 73740, M=Mean, *SD*=Standard Deviation, Med: Median, Adult: Adulthood * No M and SD were reported since Type Indicator has ordinal responses.

### The associations between demographic variables and type indicator

Kendall's tau-b (τb) correlation was performed between Type indicator score, education level, and age period. Based on the results, those with a higher level of education and within older age groups were more likely to achieve a more developmentally advanced type indicator score ranking higher on the Type indicator (Education Level; $r_\tau$= .109, $p < .001$, Age period; $r_\tau$= .053, $p < .001$). A Kruskal–Wallis H test was conducted to check for gender difference in the Type indicator score. The Kruskal–Wallis H test results indicated that there was a statistically significant difference in the Type indicator score between males and females ($\chi 2(1) = 1031.562$, $p < .001$), with a mean rank Type indicator score of 34,306.57 for males and 39,251.84 for females.

### DIT2 delivery method

We investigated whether there are gender and educational level differences in schema scores (i.e., PI, MN, and P score) and N2 summary score that are qualified by the DIT2 delivery method. Again, we used combined educational level subcategories. A series of independent 2 × 4×2 full factorial ANOVA was performed to assess whether schema and summary scores could be predicted by gender (i.e., female and male), education level (i.e., Grade 1–12, Vocational/Technical or Junior (Voc/Tech or Jr), Undergraduate, and Graduate or Professional Degree), DIT2 delivery type (online and paper) and the interaction of gender, education level, and DIT2 delivery type.

The ANOVA results showed a significant interaction effect for PI score ($F(15, 73720) = 153.402$, $p < .001$; η2 = .030), MN score ($F(15, 73720) = 57.829$, $p < .001$; η2 = .012)), P score ($F(15, 73720) = 245.999$, $p < .001$; η2 = .048)), and N2 score ($F(15, 73720) = 275.974$, $p < .001$; η2 = .053)) for gender, education level, and DIT2 delivery type. The gender and education level interaction for schema and summary scores differed for online DIT2 delivery type compared to paper DIT2 delivery type.

Simple main effects test indicated that both female and male paper DIT2 test takers received higher PI scores than online DIT2 test takers in grade 1–12 ($p < .001$ for females and males) and vocational/technical or junior education levels ($p = .002$ for females and $p < .001$ for males), while both female and male online DIT2 test takers received higher PI scores than paper DIT2 test takers in undergraduate, and graduate or professional degree levels ($p < .001$). Both female and male paper DIT2 test takers received significantly higher MN scores than online DIT2 test takers in grade 1–12 ($p < .001$ for females and males), vocational/technical or junior ($p < .001$ for females and $p = .009$ for males), undergraduate ($p < .001$ for females and males), and graduate or professional degree levels ($p < .001$ for females and males). Both female and male online DIT2 test takers received higher P scores than paper DIT2 test takers in grade 1–12 ($p < .001$ for females and $p = .009$ for males) and vocational/technical or junior education levels ($p < .001$ for females and $p = .009$ for males). In undergraduate and graduate levels, the results were non-significant. Both female and male online DIT2 test takers received higher N2 scores than paper DIT2 test takers in grade 1–12 ($p < .001$ for females and males) and vocational/technical or junior education levels ($p < .001$ for females and males). In undergraduate and graduate levels, male online DIT2 test takers received higher N2 scores than male paper DIT2 test takers ($p < .001$). The results for females in undergraduate and graduate levels were non-significant.

## DISCUSSION

Moral reasoning can be described as the mental process that a person uses to consider moral challenges and come up with potential solutions (Fisher & Ott, 1996). The DIT, as a remarkably well-validated and reliable measure, focuses on the interpretation and understanding of moral matters. Previous studies using the DIT demonstrate that moral judgment is enhanced in high school and undergraduate years. Beginning in adolescence years, youths attain awareness of discovering society (Youniss & Yates, 1997). Once they gain socio-centric perspectives, they understand societal concerns better in college (Roberts, 2020). In sophomore years, they have generally mixed levels of moral understanding. In later years in college, they can gain more advanced reasoning (Pascarella & Terenzini, 1998). The transition from high school to college is a period of increased stress because of increased responsibilities and adult roles (Rice et al., 1995), new social networks (Aquilino, 2006), financial burdens, and educational changes (Laursen & Collins, 2009). In later years, they gain intellectual, cultural, and social experiences that support the improvement of moral reasoning (Pascarella & Terenzini, 1998). The results of the DIT also showed that the measure was responsive to educational interventions, was associated with moral behaviors and decisions, and distinguished across groups of people whose moral judgment progress might reasonably be predicted to differ (Thoma & Dong, 2014). The DIT-2 (J. R. Rest, Narvaez, et al., 1999) is based on Kohlberg's study in which an interview format to gauge moral judgment was employed. Data sets preserved at the Center for the Study of Ethical Development offer a wide collection of studies that can be utilized to determine normative data regarding DIT2. As mentioned, the current article provides the norms by "education," "gender and education," and "gender and age" for the last 10 years for DIT2 schema scores (i.e., PI score, MN score, and P score) and most common summary scores (i.e., N2 score and Type indicator).

Regarding education, when mean scores for different DIT2 respondent subgroups were compared to previously published DIT2 averages, the results showed similarity for scores on the various indices for those educated at the undergraduate. However, at more advanced levels of education (e.g., Professional Degree, MS Degree, and Ph.D./Ed.D), the average P scores for the current DIT2 vs. the previous DIT2 appear to diverge. In other words, mean P scores and N2 scores of participants with a professional degree, MS degree, or Ph.D. are somewhat lower than those reported in previous DIT2 norms, but variances are similar. Societies are ever more internet connected across the world and increased time spent using digital devices might be implicated in some of the observed differences (Montag & Diefenbach, 2018). This subtle difference between DIT2 norms may also be associated with a more diverse population included in the current norms because of online DIT2 availability. In later studies, regional differences could be explored to further understanding of variance among participants with advanced education levels. In the first DIT norms, standard deviations get smaller for more educated samples, whereas variances for the more recent norms were large, suggesting great variability in participants' responses. However, in the current study, there were no major differences in the standard deviations depending on the level of education. We found certain differences between online and paper DIT2 tests depending on gender and education levels. Both female and male paper DIT2 test takers received higher PI scores and MN scores than online DIT2 test takers, especially at lower education levels. In terms of P score and N2 score, online DIT2 test takers received higher scores than paper DIT2 test takes although some results were non-significant in higher education levels. This is the first norming study where online delivery is possible, and results indicate that online or paper-pencil test-taking conditions might influence DIT2 importantly when gender and education levels are taken into account. Online DIT2 requires less processing time, is cost-efficient, easier to deliver and transmit, and faster to score. The current results are important by showing that online DIT2 might be an advantage in terms of results.

Many studies are specifically interested in the moral development of college students (J. R. Rest, 1988; King & Mayhew, 2002). Among 172 studies of college students using the DIT, 170 of them reported differences in moral reasoning of students as the result of formal education (King & Mayhew, 2002). Data from longitudinal studies also show that education plays a key role in moral reasoning development (Abdolmohammadi & Reeves, 2000; J. R. Rest & Thoma, 1985; J. R. Rest et al., 1978; Lind, 2000; Reale et al., 2018). Current DIT2 norms also support that there are significant mean differences in schema and summary scores of freshmen and senior college students, which indicates more advanced moral reasoning among seniors. Despite debates over the years about the efficacy of college ethics education, our current findings continue support for education as a possible route for advancing participants to greater levels of moral judgment (Hatton, 1996; Lopez et al., 2005). The college experience relates to significant developmental differences in moral reasoning due to several factors including course-taking patterns (Mayhew et al., 2012), types of curricular content and how college courses are presented (Armstrong, 1993; Mayhew & King, 2008), especially student-centered moral discourse (Hartwell, 1995), emphasizing diversity and social justice in the courses (Boss, 1994; Hurtado et al., 2003; Mayhew & Engberg, 2010), faculty–student interaction (J. R. Rest, 1988; Mayhew & King, 2008; McNeel, 1994), co-curricular activities (Cohen, 1982), and peer relationships (Endicott et al., 2003; Mayhew & King, 2008; Thoma & Ladewig, 1993). Spending longer in education is known also to grow social networks (Lin, 2001) and can facilitate access to and acquisition of resources to promote prosocial behavior (Wilson and Musick, 1998); these aforementioned factors may be considered as indirect effects of education on moral behavior. In a college environment, students may view diverse perspectives, gain opportunities to develop cognitive reasoning, engage in a culture that supports values of virtue and integrity (King & Mayhew, 2002), and deal with challenges for psychosocial adjustment (Conley et al., 2018).

Aside from making their own judgments on gender differences in moral reasoning, researchers have agreed that gender differences in moral reasoning are controversial (Walker, 1995). In general, it appears that both females and males exhibit and employ justice and caring orientations differently depending on context- and background-related elements (Jaffee & Hyde, 2000). Most of the fifty or so studies included in two historical meta-analyses (Thoma, 1986, 1994; Walker, 1984) examined associations between gender and moral reasoning development and indicated that college females had higher moral reasoning levels than males, somewhat contrary to charges of gender bias for Kohlbergian procedures (Mayhew et al., 2012; Walker, 2006). Regarding education, previous studies were interested in whether gender differences were more pronounced at higher levels of education (e.g., Emler et al., 1998; M. J. Bebeau, 2002). By conducting a meta-analysis of 56 DIT studies, Thoma (1986) concluded that education was 250 times more powerful than gender in accounting for the variance of DIT scores when excluding graduate students and adults in the sample. The availability of full demographic data across data sets of DIT2 helped researchers to perform broad comparisons. When all education levels were included, the previous DIT2 norms showed that differences between males and females appeared to be very modest for lower educational levels, whereas differences become more pronounced as educational level increases. In the current norms, differences in schema and summary scores between males and females by education level are also evident. In other words, this study supports earlier norms of increased differences between males and females, in particular, based on the levels of education. In all education levels, female participants scored lower on PI and MN and higher on the P score and N2. In other words, females in all education levels had higher levels of moral reasoning regardless of their education level.

Based on Kohlberg's moral judgment development theory, human development is directly linked to age (Kohlberg, 1969). Data by age group demonstrate that people change with time (J. Rest et al., 1986), and moral reasoning changes as a result of the combination of maturity and learning, which is connected to education and social experience (Piaget, 1932; Kohlberg, (1969), 1976). However, conflicting results have also been reported regarding age and moral reasoning development (Ford & Richardson, 1994). Regarding age, participants in young adulthood received higher P score, N2, and Type Indicator than participants in other age periods. The reason might be that young individuals explore a variety of behaviors in their third decade of life as they work to establish their moral identities (Lee et al., 2015), which is a process that starts in adolescence and can continue to be challenged and changed during young adulthood (Smith, 2011). Also, many young adults in our sample have college experience. As discussed earlier, young adults with college experience might use more advanced moral reasoning strategies when encountering moral dilemmas. Regarding gender and age, female participants in young adulthood received higher P score and N2 than participants in middle-late adulthood. This finding is an important contribution to the literature because research showing gender and age interaction in moral reasoning is scarce (e.g., Goolsby & Hunt, 1992). Thus, age and gender interaction were documented, suggesting that moral reasoning ability for females might be greater than for males in young adulthood. Specific to age period, some researchers stated that moral reasoning ability declines with age (e.g., Bigel, 2000; Elm & Nichols, 1993; Eynon et al., 1997). Although education and age are significant predictors of moral reasoning (J. Rest, 1986), the possibility of having fewer educational degrees in middle-late adult participants compared to young adult participants in the last 10 years DIT2 dataset could explain their lower P score and N2.

Overall, the present research provides updated general guidelines for researchers wanting to compare their data sets with a much larger sample that is diverse in terms of age and education level. Also provided is information about moral reasoning changes over time as assessed by DIT2. DIT2 has proven to be an advanced tool for the study of moral judgment and a useful resource for researchers wanting to assess moral reasoning, especially at the level of bedrock moral schemas according to neo-Kohlbergian theory.

## LIMITATIONS

The first limitation of using secondary data or establishing Norms is that we have to rely upon reports of researchers who collected the data for DIT2 and for demographic descriptions of the participants who took the test. Secondly, the norms are also limited to the last 10 years and only include datasets based on the full DIT2 (i.e., five stories) and have full demographic information (i.e., age, gender, education level, political liberalism, language, and citizenship). Future studies might be performed with datasets that include a shorter version of DIT2 (i.e., three or four stories). Thirdly, the study only includes participants whose primary language is English and who are US citizens. The DIT2 has been translated into more than 10 languages. Thus, future DIT2 norms should be designed for populations of other language speakers and non-US citizens. Also, the small effect sizes observed among educational groupings and age period groupings limit practical applications of the results. Finally, the Norms were prepared only by "education," "gender and education," and "gender and age" for DIT2 schema scores and most common summary scores. Researchers might consider conducting analyses for other summary scores and other demographic combinations in future studies.

## CONCLUSION

The current DIT2 norms include a general guideline so that researchers may make comparisons between their data sets and the larger sample reported in this paper. While comparing their data sets with normative information, researchers should consider that the norms indicate diversity in age, gender, education levels, and political liberalism.

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## DATA AVAILABILITY STATEMENT

Due to ethical concerns, supporting data cannot be made openly available. Information about the data and conditions for access are available at the Center for the Study of Ethical Development at https://ethicaldevelopment.ua.edu/

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